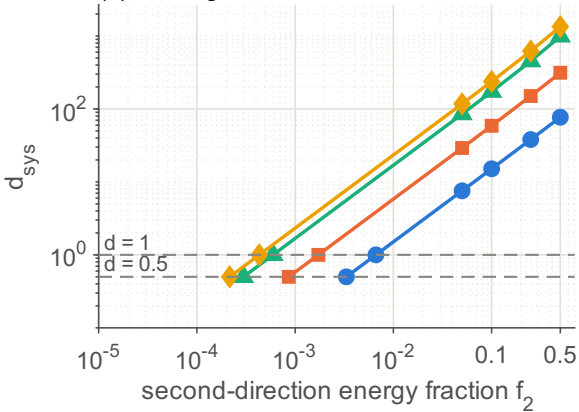
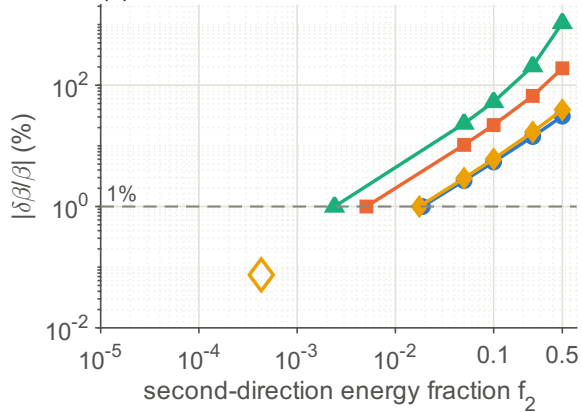


(a) Bias significance



(b) Stiffness bias



$\phi = 15^\circ$ $\phi = 30^\circ$ $\phi = 60^\circ$ $\phi = 90^\circ$ nonlinear (b)